We acknowledge the use of OpenAI's ChatGPT as a supportive tool in drafting and refining portions of the text.

Urban Ecosystems and Sustainable Landscapes (LAND 110)
Department of Environmental Systems
University of British Columbia

Course syllabus

1. Course information

Title: Urban Ecosystems and Sustainable Landscapes
Code (section): LAND 110 (201)
Term: 2026W1 (Sep 8 – Dec 4, 2026)
Delivery mode: Fully in-person
Credits: 3
Schedule: Tue - Thu, 10:00 – 11:30
Pre-requisites: None
Co-requisites: None
Syllabus version: 2026-09-03

2. Teaching team

Instructor: Dr. Melissa Chan
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Teaching Assistant: Aaron Delgado
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Teaching Assistant: Priya Raman
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3. Course description

Cities and landscapes are deeply connected systems that shape biodiversity, climate resilience, water quality, public wellbeing, and sustainability outcomes. Urban Ecosystems and Sustainable Landscapes introduces students to the relationships between environmental systems and urban development through case studies, discussions, and applied examples. Students will examine topics such as urban forests, ecological restoration, green infrastructure, stormwater systems, environmental planning, biodiversity conservation, and sustainable land-use practices. The course explores how environmental systems interact with human activity and how communities can balance development with ecological health. The course is designed for early undergraduate students interested in environmental studies, urban sustainability, planning, ecology, and resource management. LAND 110 provides students with foundational knowledge that can support future coursework across environmental and earth science disciplines.

4. Learning outcomes

General & integrative
- Explain the importance of ecological systems within urban environments.
- Identify sustainability challenges associated with urban growth and development.
- Discuss environmental planning approaches used in modern cities.

1. Urban ecology
- Describe ecological interactions in urban ecosystems.
- Explain biodiversity challenges in developed environments.

2. Sustainable infrastructure
• Identify the role of green infrastructure in sustainability planning.
• Interpret basic environmental monitoring data and indicators.

3. Environmental communication
• Evaluate environmental communication strategies for public audiences.
• Develop evidence-based arguments about sustainability challenges.

5. Course website

All updated course information will be available through the Canvas course website. Students will use Canvas to access lecture slides, assignment instructions, quizzes, readings, and announcements throughout the term.

Only registered students will have access to the course materials hosted through Canvas.

6. Course structure

6.1 In-person delivery
This course is designed for fully in-person delivery. Students are expected to attend lectures, participate in in-class activities, and engage in collaborative discussions.

6.2 Lectures
Core course concepts will be delivered during scheduled lecture periods. Lecture slides and selected readings will be uploaded to Canvas before class sessions whenever possible.

6.3 Participation activities
Participation activities and in-class engagement exercises may be completed using online polling tools, collaborative worksheets, or discussion prompts.

6.4 Discussions and workshops
Students will complete workshops and discussions connected to sustainability case studies, urban planning scenarios, and environmental communication exercises.

6.5 Midterm and final assessment
The course includes a midterm assessment and a cumulative final assessment focused on core course concepts, interpretation of case studies, and applied environmental reasoning.

6.6 Office hours
The instructional team will hold weekly office hours for questions related to course material, assignments, and assessments.

7. Learning materials

7.1 Supplies needed
• A laptop or tablet for assignments and online activities.
• Access to a mobile device for participation activities.

7.2 Textbook and readings
The course textbook is: Foster, R. & Nguyen, T. (2022). *Urban Ecology and Sustainable Landscapes: Environmental Systems in Modern Cities.* West Coast Academic Press.

Additional articles, multimedia resources, case studies, and readings will be provided through Canvas modules throughout the term.

7.3 Learning analytics
Canvas and other online participation tools may collect learning activity data to support course delivery and student

engagement.

8. Assessment of learning

Participation activities: 10%
Weekly quizzes: 12%
Workshops and discussions: 18%
Reflection assignment: 10%
Case study project: 20%
Midterm assessment: 10%
Final assessment: 20%

9. Course contents

Lecture slides are the primary source of information, and specific topics covered are indicated below:

Module	Topic	Title*
1. Urban ecology	1	Introduction to urban ecosystems
	2	Cities, biodiversity and ecological systems
	3	Urban forests and green corridors
	4	Climate resilience in cities
2. Sustainable infrastructure	5	Stormwater systems and water management
	6	Green roofs and sustainable design
	7	Transportation systems and sustainability
3. Environmental planning	8	Land-use planning and environmental policy
	9	Environmental monitoring and indicators
	10	Community engagement and urban planning
4. Communication & engagement	11	Environmental communication strategies
	12	Public outreach and sustainability campaigns
	13	Communicating scientific uncertainty
5. Professional development	P1	Research and scholarly communication
	P2	Ethics and professionalism
	P3	Collaborative problem solving

* *Guest lectures and workshops may provide additional testable material throughout the term.*

10. University policies

10.1 Academic integrity
Students are expected to submit original work and appropriately acknowledge all sources used in assignments and projects.

10.2 Student support
UBC provides academic support, counselling, accessibility services, and wellbeing resources for students throughout the academic year.

10.3 Copyright
Course materials are intended only for students enrolled in the course and may not be redistributed without permission.